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Single Qubit Quantum Systems

Multiple Qubit Quantum Systems: Direct/Tensor Product, Entanglements, Gates and Quantum Parallelism

Lab

Interactive Notes on Multi-Qubit Quantum Systems

1h

Video

Multi Qubit Quantum States

29 min

Graded Assignment

Multiple Qubit Quantum States

2h

Video

Multi Qubit Quantum Gates

28 min

Graded Assignment

Multiple Qubit Quantum Gates

40 min

Video

Quantum Parallelism

14 min

Video

No Cloning Theorem

5 min

Graded Assignment

Quantum Parallelism and Implementing Classical Gates

45 min

Bell's Inequality and An Entanglement Game

Grover's Search Algorithm

Problem Set: Week 2

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Multiple Qubit Quantum States

Next item →

Review Learning Objectives

2 / 2 points

$(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle) \otimes |1\rangle$

Assignment details

Select all the correct answers.

☐

$(\frac{1}{\sqrt{2}}|10\rangle - \frac{1}{\sqrt{2}}|11\rangle)$

☐

$|1\rangle \otimes (\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle)$

☒

$(\frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|11\rangle)$

Correct

Correct

Your grade

To pass you need at least 80%. We keep your highest score.

100%

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Like

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1. Which of the states below represents the product:

2 / 2 points

$|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$

Suppose we measured both qubits at once, what are the possible outcomes of the measurement?

☐

The second qubit collapses to either $|0\rangle$ or $|1\rangle$ with equal probability.

☒

The superposition collapses to one of the pure states $|00\rangle, |01\rangle, |10\rangle$ with equal probability $\frac{1}{3}$.

☐

The first qubit collapses to $|0\rangle$ always and the second qubit collapses to $|1\rangle$ always.

☐

The first qubit collapses to either $|0\rangle$ or $|1\rangle$ with equal probability.

Correct

Correct

2. Consider the two qubit state

3 / 3 points

$|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$

Suppose we measured just the first qubit and obtained an outcome of $|0\rangle$. What can we say about the state after the measurement?

☒

After measurement, the first qubit will be in state $|0\rangle$ whereas the second qubit will be in the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

☐

After measurement, the first qubit will be in state $|0\rangle$ whereas the second qubit will be in the state $\frac{1}{\sqrt{3}}(|0\rangle + |1\rangle)$

☐

After measurement, the first qubit will be in state $|0\rangle$ whereas the second qubit will be in the state $|1\rangle$

☐

After measurement, the first qubit will be in state $|1\rangle$ whereas the second qubit will be in the state $|0\rangle$

Correct

Correct.

3. Consider the two qubit state

2 / 2 points

$|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$

Suppose we measured just the first qubit and obtained an outcome of $|1\rangle$. What can we say about the state after the measurement?

☐

After measurement, the first qubit will be in state $|0\rangle$ whereas the second qubit will be in the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

☐

After measurement, the first qubit will be in state $|0\rangle$ whereas the second qubit will be in the state $\frac{1}{\sqrt{3}}(|0\rangle + |1\rangle)$

☐

After measurement, the first qubit will be in state $|0\rangle$ whereas the second qubit will be in the state $|1\rangle$

☒

After measurement, the first qubit will be in state $|1\rangle$ whereas the second qubit will be in the state $|0\rangle$

Correct

Correct

4. Consider the two qubit state

2 / 2 points

$|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$

Suppose we measured just the first qubit and obtained an outcome of $|1\rangle$. What can we say about the state after the measurement?

☐

After measurement, the first qubit will be in state $|0\rangle$ whereas the second qubit will be in the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

☐

After measurement, the first qubit will be in state $|0\rangle$ whereas the second qubit will be in the state $\frac{1}{\sqrt{3}}(|0\rangle + |1\rangle)$

☐

After measurement, the first qubit will be in state $|0\rangle$ whereas the second qubit will be in the state $|1\rangle$

☒

After measurement, the first qubit will be in state $|1\rangle$ whereas the second qubit will be in the state $|0\rangle$

Correct

Correct

5. Select all entangled states from the two qubit super positions below. Note that a state of a multiqubit quantum system is entangled if it **cannot be written** as a direct/tensor product of single qubit states. For instance, if $|\psi\rangle$ is a m-qubit state then

3 / 3 points

$|\psi\rangle \neq |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_m\rangle$

for any choice of single qubit superpositions $|\psi_1\rangle, \dots, |\psi_m\rangle$

☒

$\frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |11\rangle)$

Correct

Correct

☒

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Correct

Correct: we showed this in the video lecture and notes

☐

$\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$

☒

$\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$

Correct

Correct: this is an entangled state

☐

$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

https://www.coursera.org/learn/advanced-data-structures-ns-and-quantum-algo/assignment-submission/3GFu1/multiple-qubit-quantum-states/view-feedback

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